

1. NumPy arange() Program

Write a Python program using NumPy to create an array using user input (start, stop, step) and display the result using np.arange().

Code:

```
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import numpy as np

start = int(input("Enter start: "))
stop = int(input("Enter stop: "))
step = int(input("Enter step: "))

a = np.arange(start, stop, step)

print(a)
|
```

Output:

```
>>> |
===== RESTART: C:/Users/vijay/OneDrive/Documents/eds pract
Enter start: 3
Enter stop: 8
Enter step: 4
[3 7]
>>> |
>>> |
```

2. Array Operations (Arithmetic, Statistical, Bitwise)

Write a Python program using NumPy to perform arithmetic, statistical, and bitwise operations on two arrays.

Code:

```
File Edit Format Run Options Window Help
import numpy as np

def array_operations(A, B):
    A = np.array(A)
    B = np.array(B)

    # Arithmetic Operations
    sum_result = A + B
    diff_result = A - B
    prod_result = A * B

    # Statistical Operations
    mean_A = np.mean(A)
    median_A = np.median(A)
    std_dev_A = np.std(A)

    # Bitwise Operations
    and_result = A & B
    or_result = A | B
    xor_result = A ^ B

    # Print results
    print("Sum:", sum_result)
    print("Difference:", diff_result)
    print("Product:", prod_result)
    print("Mean of A:", mean_A)
    print("Median of A:", median_A)
    print("Standard Deviation of A:", std_dev_A)
    print("Bitwise AND:", and_result)
    print("Bitwise OR:", or_result)
    print("Bitwise XOR:", xor_result)

# Example
A = [1, 2, 3]
B = [4, 5, 6]

array_operations(A, B)
```

Output:

```
>>>
===== RESTART: C:/Users/vijay/OneDrive/Documents:
Sum: [5 7 9]
Difference: [-3 -3 -3]
Product: [ 4 10 18]
Mean of A: 2.0
Median of A: 2.0
Standard Deviation of A: 0.816496580927726
Bitwise AND: [0 0 2]
Bitwise OR: [5 7 7]
Bitwise XOR: [5 7 5]
>>>
```

3. View vs Copy in NumPy

Write a Python program to demonstrate the difference between a view and a copy of a NumPy array.

Code:

```
import numpy as np
inputlist = list(map(int, input().split()))
original_array = np.array(inputlist)
# Create a view
view_array = original_array.view()
# Create a copy
copy_array = original_array.copy()
# Modify the view
view_array[0] = 99
print("Original array after modifying view:", original_array)
print("View array:", view_array)
# Modify the copy
copy_array[1] = 88
print("Original array after modifying copy:", original_array)
print("Copy array:", copy_array)
```

Output:

```

> -----
> ===== RESTART: C:/Users/vijay/OneDrive/Documents/eds practical.py =====
1 2 3 4
Original array after modifying view: [99  2  3  4]
View array: [99  2  3  4]
Original array after modifying copy: [99  2  3  4]
Copy array: [ 1 88  3  4]
> |

```

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4. Searching, Counting, Broadcasting, Sorting

Write a Python program using NumPy to perform searching, counting, broadcasting, and sorting on an array.

Code:

```

import numpy as np

# Input array
array1 = np.array(list(map(int, input().split()))))

# Searching
search_value = int(input("Value to search: "))
count_value = int(input("Value to count: "))
broadcast_value = int(input("Value to add: "))

# Find indices
a = np.where(array1 == search_value)[0]
print(a)

# Count occurrences
b = np.count_nonzero(array1 == count_value)
print(b)

# Broadcasting addition
c = array1 + broadcast_value
print(c)

# Sorting
d = np.sort(array1)
print(d)

```

Output:

```
===== RESTART: C:/Users/vijay/OneDriv
1 2 3 4 2 1
Value to search: 5
Value to count: 2
Value to add: 8
[]
2
[ 9 10 11 12 10  9]
[1 1 2 2 3 4]
|
```